Microcontroller-based Voltage Protector Device

1. Design Approach

The device protects appliances from grid voltage surges (over-voltage), under-voltage, and over-current faults. An Arduino Uno reads two analog channels: one potentiometer scales 0–5 V ADC to a simulated 0–300 V AC range, the second maps to 0–10 A load current. Protection thresholds are compared in software. When a fault is detected the microcontroller de-energises a relay, disconnecting the load. A 16×2 LCD and three buttons provide local configuration; a 9600-baud UART interface allows a computer to query status and change thresholds in real time. All sensing uses low-voltage stand-ins only – no mains connection is required or recommended.

2. Implementation

Hardware (simulation): Arduino Uno, two potentiometers (voltage & current), 5 V relay module, 16×2 LCD (4-bit mode), three push-buttons (INPUT_PULLUP), optional buzzer, LED/lamp as load.

Firmware: Continuous ADC sampling, linear scaling, three-threshold comparison with hysteresis for auto-reconnect, non-blocking button debounce, menu-driven LCD (Normal / Set-OV / Set-UV / Set-OC), and a simple text protocol (STATUS, SET_OV, SET_UV, SET_OC, CONNECT, DISCONNECT).

Set Parametrs: Over-voltage 250 V, Under-voltage 180 V, Over-current 5.0 A – all adjustable at runtime.

3. Results & Behaviour

In simulation the system correctly trips the relay within one ADC cycle when any threshold is exceeded. The LCD immediately shows the fault type (OVER VOLTAGE / UNDER VOLTAGE / OVER CURRENT). Returning the potentiometers inside the safe window (with 5 V / 0.2 A hysteresis) automatically reconnects the load. Serial commands update thresholds instantly and STATUS reports live readings. Local button navigation works without interfering with the protection loop. The design is intentionally simple and transparent so that behaviour under evaluation is predictable and easy to verify.

4. What Was Completed / Limitations

- Full protection logic (OV / UV / OC) with hysteresis and auto-reconnect
- Dual user interfaces: onboard LCD+buttons and computer serial protocol
- Complete, well-commented Arduino source and documentation
- Simulation-oriented sensing (pots instead of real PT / CT) – intentional for safety

Limitation: No physical hardware was built; verification is by code review and simulation Wokwi. A real voltage sensor (e.g. ZMPT101B) and current sensor (ACS712) would replace the pots for hardware deployment.

5. Repository & How to Reproduce

Source: src/voltage_protector.ino **Report:** this document **Full instructions:** readme.md Open the .ino in Arduino IDE Wokwi , select Arduino Uno, start simulation, open Serial Monitor at 9600 baud, and exercise the pots and buttons as described in the readme.